

SS1 Biology Diagnostic (Part 2)

Curriculum: Cambridge IGCSE / O-Level

Cluster: Ecology, Genetics, Reproduction & Environment (Questions 21–40)

No.	Question	Ans	Diagnostic Focus
21	The process by which green plants lose water vapour is: A) Respiration B) Transpiration C) Diffusion D) Guttation	B	Plant physiology
22	Which pigment traps light energy for photosynthesis? A) Haemoglobin B) Chlorophyll C) Melanin D) Keratin	B	Photosynthesis
23	An organism that can manufacture its own food is called a: A) Consumer B) Decomposer C) Producer D) Parasite	C	Ecology
24	The sequence of organisms through which energy passes is a: A) Food web B) Food chain C) Habitat D) Niche	B	Food relationships
25	Bacteria and fungi are mainly: A) Producers B) Consumers C) Decomposers D) Predators	C	Nutrient cycling
26	The basic unit of heredity is the: A) Chromosome B) Gene C) Cell D) Tissue	B	Genetics
27	A normal human body cell contains: A) 23 chromosomes B) 46 chromosomes C) 44 chromosomes D) 92 chromosomes	B	Chromosome number
28	The genotype 'Aa' is described as: A) Homozygous dominant B) Homozygous recessive C) Heterozygous D) Dominant only	C	Inheritance
29	The physical expression of a gene is the: A) Genotype B) Allele C) Phenotype D) Chromatid	C	Genetics vocabulary
30	Which cell division produces gametes? A) Mitosis B) Binary fission C) Meiosis D) Budding	C	Cell division
31	Fertilization is the fusion of: A) Two body cells B) Two gametes C) Two tissues D) Two embryos	B	Reproduction
32	The male reproductive organ in flowering plants is the: A) Pistil B) Ovule C) Anther D) Ovary	C	Plant reproduction
33	Which disease is caused by a virus? A) Tuberculosis B) Cholera C) Measles D) Malaria	C	Human health
34	Vaccination works by stimulating the production of: A) Enzymes B) Hormones C) Antibodies D) Vitamins	C	Immunity
35		B	Environmental biology

	Excessive cutting of trees leads directly to: A) Afforestation B) Deforestation C) Succession D) Pollination		
36	Which gas is mainly responsible for the greenhouse effect? A) Oxygen B) Nitrogen C) Carbon dioxide D) Neon	C	Climate science
37	The variety of living organisms in an ecosystem is called: A) Biomass B) Biodiversity C) Population D) Habitat	B	Conservation
38	Which practice best conserves biodiversity? A) Bush burning B) Overgrazing C) Establishing protected areas D) Excessive logging	C	Environmental management
39	Natural selection was proposed by: A) Gregor Mendel B) Charles Darwin C) Louis Pasteur D) Robert Hooke	B	Evolution
40	Diagnostic Insight: Students who perform strongly on genetics, ecology and environmental biology questions demonstrate readiness for advanced Biology topics in SS1 and beyond.	—	Overall readiness